REDUCED LOCALLY NOETHERIAN ONE-POINT STACKS ARE REGULAR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3033-3037 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 3039-3051, the complete author proof (11 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 26 of queue_stacks-properties.json (pb=655, stmt 3033-3037, proof 3039-3051).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-unique-point-better: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2710–2717. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06MK: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 3051–3051. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06M1: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 3051–3051. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *A reduced, locally Noetherian algebraic stack $\mathcal{Z}$ such that $|\mathcal{Z}|$ is a singleton is regular.*

Proof. Let $W \to \mathcal{Z}$ be a surjective smooth morphism where W is a scheme. Let k be a field and let $\mathrm{Spec}(k) \to \mathcal{Z}$ be surjective, flat, and locally of finite presentation (see Lemma lemma-unique-point-better). The algebraic space $T = W \times_{\mathcal{Z}} \mathrm{Spec}(k)$ is smooth over k in particular regular, see Spaces over Fields, Lemma 06M1. Since $T \to W$ is locally of finite presentation, flat, and surjective it follows that W is

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regular, see Descent on Spaces, Lemma 06MK. By definition this means that $\mathcal{Z}$ is regular. $\square$